MATLAB / SIMULINK PROJECT

Multi-Variable Boiler Automation System

PID-Based Coupled Control in MATLAB / Simulink

Control Loops	PID Controllers	Process Variables	Safety Alarms
2	2	2+	1

Industrial Automation | Process Control | DCS / SCADA Simulation

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Abstract

This project presents the design and simulation of a **Multi-Variable Boiler Automation System** using **MATLAB/Simulink**. The objective is to simulate an industrial boiler control system capable of maintaining stable **water level** and **temperature** using PID-based closed-loop control. The system includes disturbance rejection, coupled process interaction, real-time monitoring, and safety alarm mechanisms. The project demonstrates how industrial automation systems such as **Distributed Control Systems (DCS)** and **SCADA** platforms maintain safe and efficient boiler operation.

1. Introduction

Industrial boilers are critical components in power plants, manufacturing industries, chemical plants, and process industries. Boilers generate steam by heating water and therefore require continuous monitoring and control to ensure safe and efficient operation.

The two most important variables in boiler operation are **Boiler Water Level** and **Boiler Temperature**. Improper control of either variable can result in reduced efficiency, equipment damage, or unsafe operating conditions. Therefore, industrial control systems use feedback controllers such as **PID controllers** to maintain these variables at their desired set-points.

This project simulates such an industrial boiler automation system using MATLAB/Simulink, providing a practical demonstration of feedback control, process coupling, disturbance rejection, and SCADA-style monitoring.

2. Project Objectives

- Simulate an industrial boiler system in MATLAB/Simulink.
- Implement PID-based closed-loop control for multi-variable regulation.
- Maintain stable boiler **water level** under varying conditions.
- Maintain stable boiler **temperature** with realistic thermal dynamics.
- Simulate industrial disturbances and demonstrate rejection capability.
- Demonstrate interaction (coupling) between process variables.
- Implement a safety monitoring and alarm system for over-temperature detection.
- Create a SCADA/DCS-style real-time monitoring dashboard.

3. System Overview

The project consists of two major closed-loop control systems that are coupled through a gain-based interaction path, replicating the interdependence found in real industrial boilers.

Control Loop	Controlled Variable	Actuator	Sensor / Feedback
Water Level Loop	Boiler Water Level	Feed-water Valve	Level Transmitter
Temperature Loop	Boiler Temperature	Burner / Heater	Temperature Sensor

Unlike independent single-loop systems, both loops are interconnected through a **coupling mechanism** that simulates real industrial process interaction — where water level changes influence temperature dynamics and vice versa.

4. Water Level Control Loop

The water level control loop is responsible for maintaining the desired water level inside the boiler. It forms the primary control loop and also provides the coupling signal to the temperature loop.

4.1 Simulink Components

Component	Description
Step Input	Water Level Setpoint — desired water level value
Sum Block	Computes error: Error = Setpoint – Actual Output
PID Controller	Generates corrective control signal from error
Disturbance Input	Sine-wave signal simulating load/pressure fluctuations
Boiler Dynamics TF	Transfer function modelling physical boiler behaviour
Feedback Path	Routes actual output back to the Sum block
Gauge & Display	Real-time visualisation of water level
Alarm System	Monitors level limits and triggers alarms

4.2 Working Principle

1. The operator specifies a desired water level via the **Step Input** block.

2. Actual water level is continuously fed back to the **Sum block**, which calculates: *Error = Setpoint – Actual Output*.

3. The **PID Controller** processes the error and generates a corrective control signal.

4. The control signal passes through the **Boiler Dynamics Transfer Function**, introducing realistic process delay and inertia.

5. The resulting output is fed back, forming a **closed-loop** that automatically corrects deviations.

5. Temperature Control Loop

The temperature control loop regulates boiler temperature to its desired operating set-point. It accounts for the slower thermal dynamics of the boiler and receives a coupling signal from the water level loop.

5.1 Simulink Components

Component	Description
Temperature Setpoint	Desired operating temperature input
Error Calculation Block	Computes temperature error
Temperature PID Controller	Generates heating control signal
Coupled Interaction Block	Receives gain-scaled water level signal
Temperature Dynamics TF	1/(8s + 1) — thermal process model
Gauge & Display	Real-time temperature visualisation
Over-temperature Detection	Compare block: Temperature > 1.2
Alarm Lamp	Visual alarm indicator on dashboard

5.2 Working Principle

1. The **temperature setpoint** represents the desired operating temperature.

2. Actual temperature is continuously compared with the setpoint to generate an error signal.

3. The **Temperature PID Controller** produces the required heating control signal.

4. The signal passes through the **temperature transfer function** 1/(8s+1), modelling slow thermal response.

5. A **coupling gain** from the water level loop modifies the effective thermal load.

6. Continuous feedback ensures stable, accurate temperature regulation.

6. PID Controller Design

The **PID (Proportional-Integral-Derivative)** controller is the primary decision-making element in both control loops. Its three-term architecture provides fast, accurate, and stable control across a wide range of operating conditions.

Term	Symbol	Function	Effect on Control
Proportional	K■	Responds to current error magnitude	Fast initial response; may leave steady-state error
Integral	K■	Accumulates error over time	Eliminates steady-state error; improves accuracy
Derivative	K■	Responds to rate of error change	Predicts future error; improves stability & damping

6.1 PID Control Law

u(t) = K■·e(t) + K■·∫e(t)dt + K■·de(t)/dt

Where: **u(t)** = control output, **e(t)** = error signal = Setpoint – Actual, **K■** = Proportional gain, **K■** = Integral gain, **K■** = Derivative gain

6.2 Tuning Considerations

In this project two separate PID controllers are implemented — one for each control loop. The water level PID is tuned for a faster process response, while the temperature PID is tuned for the slower thermal dynamics characterised by the larger time constant (8 s) in the transfer function. Proper tuning ensures minimal overshoot, fast settling time, and robust disturbance rejection.

7. Boiler Dynamics — Transfer Functions

The physical behaviour of the boiler is represented mathematically using **transfer function blocks** in Simulink. These introduce realistic process characteristics such as time delay, inertia, and gradual settling behaviour.

7.1 Water Level Dynamics

The water level was modelled using a **first-order or second-order transfer function** representing the hydraulic filling/draining dynamics of the boiler vessel. The transfer function introduces process delay and prevents instantaneous response, closely matching real feed-water control behaviour.

Water Level TF: $G_L(s) = K / (Ts + 1)$ — First-order process model

Where **K** = process gain, **T** = time constant (s)

7.2 Temperature Dynamics

The temperature loop was modelled with a first-order transfer function having a **larger time constant** to represent the slower thermal response of industrial boilers. Temperature changes require significant energy transfer and therefore respond more slowly than water level changes.

Temperature TF: $G_T(s) = 1 / (8s + 1)$

Time constant = **8 seconds** — models slow thermal inertia of the boiler

7.3 Importance of Transfer Functions

- Allow **mathematical modelling** of real industrial processes without physical hardware.
- Introduce **realistic dynamics** — delay, overshoot, and settling behaviour.
- Enable **frequency-domain analysis** and controller tuning.
- Make the simulation **representative of actual DCS/SCADA implementations**.

8. Disturbance Rejection

Industrial systems rarely operate under ideal conditions. Real boilers experience continuous disturbances due to varying steam demand, pressure fluctuations, and process uncertainty. This section details how the system handles such conditions.

8.1 Disturbance Implementation

A **sine-wave disturbance signal** was injected directly into the water level control loop to simulate real-world operating perturbations. The disturbance continuously attempts to push the controlled variable away from its set-point.

8.2 Types of Disturbances Modelled

Disturbance Type	Physical Representation
Load Variations	Changes in steam consumption by downstream processes
Steam Demand Changes	Sudden increases or decreases in steam output requirements

Pressure Fluctuations	Variations in boiler pressure affecting water level
Process Uncertainty	Unmeasured variables and modelling inaccuracies

8.3 PID Disturbance Rejection

The PID controller continuously monitors the error between set-point and actual output. When a disturbance shifts the process variable away from the desired value, the **Integral term** accumulates the offset and drives corrective action, while the **Derivative term** anticipates the rate of change and applies damping. Together, these mechanisms restore stable operation quickly and efficiently.

9. Coupled Multi-Variable Control

One of the most important and distinguishing features of this project is the implementation of **process coupling** between the water level and temperature loops.

9.1 What is Process Coupling?

Coupling occurs when one process variable directly influences another process variable. In real industrial boilers, the quantity of water inside the vessel directly affects the energy required to reach and maintain target temperature.

9.2 Physical Basis

- **More water** → Higher thermal mass → **More energy required** to heat → Slower temperature rise
- **Less water** → Lower thermal mass → **Less energy required** → Faster temperature response
- Water level changes therefore **directly shift the temperature dynamics**.

9.3 Simulink Implementation

Coupling Path: Water Level Output → Gain Block → Temperature Loop Input

A **Gain block** scales the water level output signal and feeds it into the temperature control loop, modelling the physical interaction between liquid volume and thermal dynamics.

9.4 Advantages of Coupled Modelling

- Simulates **realistic industrial boiler behaviour** more accurately than decoupled models.
- Demonstrates **interacting process control** — a key challenge in industrial automation.
- Makes the model **consistent with actual DCS implementations** where interaction is inherent.
- Requires more sophisticated PID tuning, better representing **real-world engineering challenges**.

10. Safety Monitoring & Alarm System

Industrial automation systems must continuously monitor safety conditions and alert operators when process variables exceed safe operating limits. An **over-temperature alarm system** was implemented to simulate this critical functionality.

10.1 Alarm System Components

Component	Role	Trigger Condition
Compare Block	Monitors temperature continuously	Temperature > 1.2 (normalised)
Alarm Lamp	Visual indicator on dashboard	Activates when Compare = TRUE
Boolean Output	Alarm state flag	TRUE = unsafe, FALSE = safe

10.2 Alarm Logic

Alarm Condition: IF Temperature > 1.2 THEN Alarm = TRUE

The Compare block evaluates this condition at every simulation time-step. When the threshold is breached, the alarm output transitions to **TRUE** and the **Alarm Lamp** on the dashboard changes state, signalling an unsafe condition to the operator — replicating the behaviour of industrial safety interlocks.

11. SCADA / DCS Monitoring Dashboard

A comprehensive monitoring dashboard was created using **Simulink Dashboard blocks** to simulate the operator interface found in modern industrial control rooms, DCS platforms, and SCADA systems.

11.1 Dashboard Components

- **Gauges** — Analogue-style dial displays for water level and temperature — providing instant visual status
- **Numeric Displays** — Precise digital readouts of current process variable values
- **Alarm Indicators** — Colour-change lamp indicators that activate on unsafe conditions
- **Set-point References** — Visual indicators of desired operating targets

11.2 Industrial Significance

The dashboard replicates the operator interface commonly found in industrial environments. Real DCS and SCADA systems provide operators with similar real-time visualisation, alarm management, and process trending capabilities. By incorporating these elements, the project demonstrates an end-to-end industrial automation workflow — from process measurement to control action to operator notification.

12. Simulation Results

The MATLAB/Simulink simulation was executed and the following outcomes were observed across all system components.

✔ Water Level Control

- Successful stable regulation of water level to desired set-point.
- Effective rejection of sine-wave disturbance — level returned to set-point after perturbation.
- Closed-loop feedback demonstrated rapid error correction.
- PID integral action eliminated steady-state offset.

✔ Temperature Control

- Stable temperature regulation achieved with the 1/(8s+1) transfer function.
- Realistic slower thermal response observed — consistent with physical boiler behaviour.
- Coupled interaction from water level successfully modelled and compensated.
- Derivative action provided smooth, well-damped temperature response.

✔ Coupled Behaviour

- Changes in water level were observed to influence temperature dynamics through the gain-coupling path.
- Multi-variable interaction demonstrated realistically within the simulation environment.

✔ Safety Monitoring

- Over-temperature alarm triggered correctly when temperature exceeded the 1.2 threshold.
- Alarm lamp transitioned to active state, confirming correct safety logic implementation.

13. Applications

The control concepts, methodologies, and simulation techniques demonstrated in this project are directly applicable to a wide range of real-world industrial systems.

Industry / Domain	Application	Relevant Concepts Used
Thermal Power Plants	Boiler drum level and steam temperature control	PID, disturbance rejection, multi-loop

Chemical Process Plants	Reactor temperature and feed flow control	Coupled control, transfer functions
Steam Generation Systems	Industrial process steam production	Level control, safety alarms
Oil & Gas Industry	Separator pressure and temperature regulation	Multi-variable PID, SCADA
Food & Beverage	Pasteurisation temperature control	Thermal dynamics, PID tuning
Pharmaceutical	Autoclave sterilisation systems	Precise temperature control, alarms
DCS / SCADA Platforms	Real-time monitoring and automated control	Dashboard, alarm management

14. Conclusion

This project successfully implemented a **Multi-Variable Boiler Automation System** using MATLAB/Simulink, demonstrating PID-based closed-loop control for simultaneous regulation of boiler water level and temperature.

Key Achievements:

- **Dual PID Control:** Two independent yet coupled PID controllers maintained stable regulation of both process variables.
- **Realistic Process Modelling:** Transfer functions introduced authentic boiler dynamics including time delays and thermal inertia.
- **Disturbance Rejection:** The control system successfully compensated for continuous sine-wave disturbances.
- **Multi-Variable Coupling:** Water level influence on temperature was modelled, demonstrating realistic process interaction.
- **Safety Systems:** Over-temperature alarm detection replicated industrial safety interlock behaviour.
- **SCADA Dashboard:** Real-time operator monitoring interface with gauges, displays, and alarm indicators.

The project provides a **realistic, complete simulation** of a modern industrial boiler automation system and highlights the critical importance of PID control, process coupling awareness, and safety monitoring in industrial applications. The methodologies demonstrated are directly transferable to real DCS/SCADA implementations in power generation, chemical processing, and manufacturing industries.

Technologies & Tools Used:

Tool / Technology	Purpose

MATLAB	Numerical computation, signal processing, analysis
Simulink	Block-diagram modelling and simulation
Simulink PID Controller Block	Closed-loop feedback control implementation
Simulink Transfer Function Block	Process dynamics modelling
Simulink Dashboard Blocks	SCADA-style operator monitoring interface
Simulink Signal Routing	Coupling path and disturbance injection

This project demonstrates the foundational principles of industrial process control and automation — applicable to any engineer working with DCS, SCADA, or PLC-based systems.